

StarBiochem

Name_____

Lysozyme Exercise

Background

In this exercise, you will use StarBiochem, a protein 3-D viewer, to explore the structure of the protein lysozyme. Lysozyme degrades the cell walls of bacteria, and is found in human tears.

Getting started with StarBiochem

- To get to StarBiochem, please navigate to: <http://web.mit.edu/star/biochem>.
- Click on the **Start** button to launch the application.
- Click **Trust** when a prompt appears asking if you trust the certificate.
- In the top menu under **Import** click on **RCSB (Protein Data Bank)** and type “1GXX” in the search window. Select “1GXX” from the list and click **Open**.

You are now viewing the structure of lysozyme, with each bond in the protein drawn as a line.

Take a moment to look at the structure from various angles by rotating and zooming on the structure.

- Instructions for changing the view of structure can be found in the top menu, under **View** → **Structure viewing instruction**.

The current way you are viewing the structure is by seeing each atom and bond in the protein drawn as a ball and a line, respectively. This way of representing a structure is called the **ball-and-stick** model and is the default model in StarBiochem. The **ball-and-stick** model allows you to see how atoms in the structure bond together. However, the space each atom occupies IS NOT accurately represented. To see a more realistic representation of the atoms in the structure you can use the **space-filled** model, where each atom is drawn as a sphere, whose size represents the physical space an atom occupies.

You can switch from the **ball-and-stick** model to the **space-filled** model in StarBiochem by increasing the size of the atoms in the structure:

- Notice that different atoms are slightly different in size.
- Gray = Carbon, Blue = Nitrogen, Red = Oxygen, Yellow = Sulfur
- Click on the **Primary** tab. The default atom size is 20% (**ball-and-stick** model).
- Move the **Atoms Size** slider to 100% (**space-filled** model).

The last page, Reference page, contains a series of terms and useful information that you will refer to during this exercise.

Exercise

Protein Structure Questions

1 Other than hydrogen, proteins contain four additional atoms, Carbon, Nitrogen, Oxygen, and Sulfur. *Which of these four atoms is most rarely found in proteins?*

Answer

2 *How many of the sulfurs in the protein are clearly exposed on the surface of this protein?*

Answer

3 Now explore lysozyme's secondary structure.

a) *How many helices are in one lysozyme protein?*

Answer

b) *Which structure predominates in the lysozyme protein, helical shapes or sheet-like shapes?*

Answer

4 *Which are more common in lysozyme, acidic amino acids or uncharged polar amino acids?*

Answer

5 Lysozyme consists of 129 amino acids. *List in order the 12 amino acids numbered 4 through 15 in lysozyme.*

Answer

6 These 12 amino acids make up a helix in lysozyme. *How many full turns of the helix (i.e. 360° turns) do you see in this one helix?*

Answer

7 Every amino acid is made up of two parts, the backbone (which is shared by all amino acids, and consists of an amino group bound to a central carbon bound to a carboxyl group) and a side chain (which is different for each of the 20 amino acids). ***Do the side chains of the amino acids in a helix point INTO or OUT of the center of the helix?***

Answer

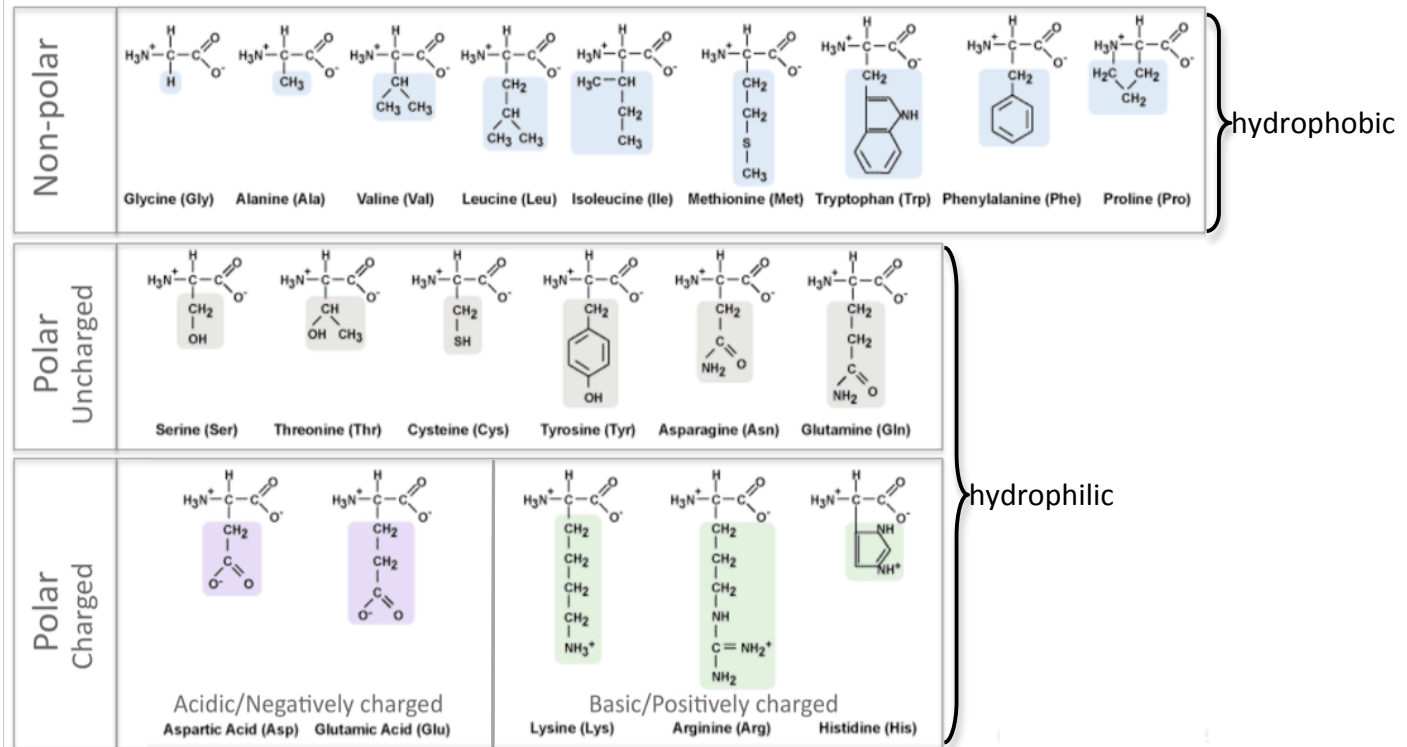
8 ***How many prolines are in one lysozyme protein?***

Answer

Reference

CHEMICAL STRUCTURES OF THE AMINO ACIDS

The 20 amino acids share a common backbone and are distinguished by different side chains, also called 'R' groups, highlighted by the various colors below.



PROTEIN STRUCTURE BASICS

All proteins have the following three levels of protein structure:

Primary structure

Describes the order of the amino acids in the protein chain but does not describe its shape.

Secondary structure

Describes shapes that form from local folding of regions within the amino acid chain. These smaller structures can be divided into two main types: helices and sheets. Coils are made of amino acids that do not form regular secondary structures (helices and sheets) but play important roles in protein folding.

Tertiary structure

Describes the entire folded shape of a protein chain.

In addition, some proteins interact with themselves or with other proteins to form larger protein structures. These proteins have an additional level of protein structure:

Quaternary structure

Describes how multiple protein chains interact and fold to form a larger protein complex.